MedFedChain: An Adaptive Differential Privacy and Blockchain-Orchestrated Federated Learning Framework for Secure Multi-Institutional Electronic Health Record Analytics *

Samia Sayed[1], Muammar Shahrear Famous[1], Rashed Mazumder[1], Risala T. Khan[1], M. Shamim Kaiser[1], Mohammad Shahadat Hossain[2], and Karl Andersson[3]

[1] IIT, Jahangirnagar University, Savar, Dhaka, Bangladesh
{samia.stu20173, shahrear.stu2017, rmiit, risala, mskaiser}@juniv.edu
[2] CSE, University of Chittagong, Chittagong, Bangladesh
hossain_ms@cu.ac.bd
[3] CSESE, Luleå University of Technology, Skellefteå, Sweden
karl.andersson@ltu.se

Abstract

Electronic Health Records (EHRs) represent one of the most sensitive and valuable assets in modern healthcare. Despite widespread digitalisation, EHR systems remain fragmented across clinical institutions, impeding collaborative diagnostics and population-level analytics while raising significant privacy and security concerns. Existing federated learning (FL) approaches address data siloing but remain vulnerable to gradient inversion attacks and lack transparent auditability. Blockchain-based solutions provide immutability and decentralisation yet introduce latency and scalability bottlenecks when used for direct data handling. The present work introduces MedFedChain, an innovative framework that integrates: (i) the adaptive differential privacy (ADP) process that calibrates noise injection based on per-layer gradient sensitivity; (ii) an InterPlanetary File System (IPFS)-based encrypted model repository for efficient off-chain storage of FL model updates; and (iii) Ethereum smart contracts for decentralised, tamper-proof aggregation and incentivisation across federated hospital nodes. Extensive experiments on two recent, large-scale public healthcare datasets—the Kaggle Diabetes Prediction (2023, 100,000 records) Dataset and the CDC BRFSS Diabetes Health Indicators Dataset (2021, 253,680 records, CDC/UCI)—demonstrate that MedFedChain achieves a classification accuracy of 91.8% with a privacy budget of $\varepsilon = 1.0$, outperforming FedAvg (85.4%), FedProx (87.1%), DP-FedAvg (88.3%), and EPP-BCFL (89.6%) whilst reducing communication overhead by 34.6%. Security analysis confirms resistance to membership inference and model poisoning attacks. These results demonstrate that MedFedChain offers a compelling and privacy-compliant foundation for collaborative EHR analytics, achieving strong performance across classification accuracy, communication efficiency, and security resistance.

1 Introduction

The expansion of Internet of Medical Things devices, coupled with accelerating clinical digitalisation, has resulted in an unprecedented volume of Electronic Health Records (EHR). According to the World Health Organization, more than 95% of hospitals in developed nations have adopted EHR systems as of 2024, with global EHR data volumes projected to surpass 2.3 zettabytes by 2028 [1]. Despite

*Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. 50, July 8-10, 2026, Seoul, South Korea.
© The copyright of this paper remains with the author(s).

this abundance, EHR data remains highly compartmentalised, locked within individual institutions by strict privacy legislation—including HIPAA in the USA, GDPR in the EU, and analogous frameworks elsewhere—creating persistent "data silos" that inhibit cross-institutional analytics and collaborative AI model training [2].

Centralised data aggregation, historically the default strategy for training diagnostic AI models, is no longer viable at scale. Pooling raw patient records across hospitals exposes sensitive information to data breaches, re-identification attacks, and regulatory violations [3]. The consequences are substantial: the average healthcare data breach in 2024 cost an estimated USD 10.9 million per incident [4]. Federated Learning (FL), first formalised by McMahan et al. [5] and extended for heterogeneous settings by Li et al. [6], offers a compelling alternative: participating institutions train local models on their private dataset and share model parameter updates only—not any raw records—with the central aggregation server. This decentralised paradigm preserves data locality whilst enabling collaborative generalisation.

Nevertheless, FL alone is insufficient for healthcare applications. Research has demonstrated that raw gradient exchanges stay mostly vulnerable to gradient inversion & membership inference attacks; through which malicious actors can partially rebuild private training samples [7, 8]. Furthermore, centralised aggregation servers represent single points of failure and trust—an unacceptable assumption in competitive multi-hospital environments. Differential Privacy (DP), formally synthesised for modern federated contexts by Ouadrhiri et al. [9], provides formal, mathematically provable privacy preservation through the injection of calibrated Gaussian noise into model updates before transmission. However, existing DP-FL approaches universally apply a static noise budget, ignoring the heterogeneous sensitivity of different model layers and clinical data distributions, thus suboptimally trading privacy for accuracy [10, 11].

Blockchain has come as a promising foundation for decentralised trust in FL systems [12, 13]. Smart contracts can automate model aggregation, enforce participation rules, and maintain an immutable audit trail—eliminating the need for a trusted central server. Prior work combining blockchain with FL in healthcare has demonstrated substantial promise [14, 15, 16] yet suffers from a critical limitation: storing large model parameter tensors directly on-chain is prohibitively expensive and slow. The InterPlanetary File System, a content-addressed peer-to-peer storage protocol, resolves this bottleneck by enabling off-chain storage of encrypted model updates, relying solely on cryptographic content hash identifiers recorded on-chain [17].

Despite significant progress in federated learning, differential privacy, and blockchain-based healthcare systems, existing approaches fail to simultaneously address privacy leakage, scalability limitations, and fair participation among heterogeneous institutions. Federated learning alone remains vulnerable to gradient-based attacks, while differential privacy often introduces a trade-off between data utility and protection. Similarly, blockchain-based solutions improve trust and auditability but suffer from high storage and computational overhead. Therefore, designing a unified system that simultaneously ensures privacy, scalability, and fairness in multi-institutional EHR analytics remains an open challenge. TThis study provides the following key contributions:

- We introduced MedFedChain, a unified framework that integrates federated learning, adaptive differential privacy, blockchain orchestration, and IPFS-based off-chain storage for secure EHR analytics.
- We introduce an Adaptive Differential Privacy (ADP) mechanism which can dynamically adjusts noise at the layer level based on gradient sensitivity, improving the privacy–utility baclance compared to static DP approaches.
- We design a blockchain-based coordination and incentive mechanism using Ethereum smart contracts, enabling decentralised aggregation, transparent participation, and reputation-based contribution weighting.

- We implement an IPFS-based off-chain storage strategy for encrypted model updates, significantly reducing on-chain storage overhead and improving system scalability.
- We conduct comprehensive experiments on large-scale healthcare datasets, demonstrating improved accuracy, reduced communication overhead, and enhanced robustness against privacy and security attacks compared to existing baselines.

The remainder of this paper is structured as follows. Section 2 surveys related work across FL, DP, blockchain, and IPFS in healthcare contexts. Section 3 details the MedFedChain architecture, ADP mechanism, smart contract design, and IPFS integration protocol. Section 4 presents experimental setup, datasets, baselines, and results with comparative analysis. Section 5 concludes and outlines future directions.

2 Literature Review

This section surveys the state of the art across the four intersecting domains underpinning this work: federated learning in healthcare (Section 2.1), differential privacy for FL (Section 2.2), blockchain-enabled healthcare systems (Section 2.3), and IPFS for decentralised medical data (Section 2.4). Table 1 synthesises key properties of representative recent works.

2.1 Federated Learning in Healthcare

McMahan et al. [5] established the theoretical foundation for FL with their FedAvg algorithm, demonstrating communication-efficient training of deep neural networks across decentralised data, but does not address privacy leakage. Subsequent work by Li et al. [6] introduced FedProx, adding a proximal regularisation term to stabilise convergence under data heterogeneity—a prevalent challenge in healthcare where patient demographics, recording devices, and clinical workflows vary substantially between sites.

In the clinical domain, Rieke et al. [2] provided a comprehensive vision for FL in medicine, identifying opportunities in radiology, genomics, and EHR analytics. Sheller et al. [18] demonstrated that FL achieves competitive performance to centralised training in brain tumour segmentation across 10 institutions, with only marginal accuracy degradation. More recently, Shah et al. [19] applied FL to EHR mortality prediction using MIMIC-III, achieving an AUC of 0.912 with FedAvg whilst preserving full data privacy—validating FL's practical applicability to tabular clinical data.

Nguyen et al. [20] introduced FedHealth, a transfer-learning-augmented FL framework specifically targeting IoMT wearable data for chronic disease monitoring. Their system demonstrated robust performance under non-IID conditions, improving accuracy by 6.3% over vanilla FedAvg. However, none of these works addressed the security of the aggregation channel, leaving gradients exposed during transit.

The landscape of trustworthy FL has expanded considerably in 2024–2025. Madupati et al. [21] proposed FL pipelines combined with blockchain-based auditing for decentralised data pipeline accountability, demonstrating improved trustworthiness but encountering scalability limits with large hospital networks. Pavlova et al. [22] introduced a decentralised FL environment leveraging smart contracts and verifiable secret sharing for open-source model training, advancing cryptographic security at the cost of substantially higher training overhead. Similarly, Mu et al. [23] addressed medical image heterogeneity across institutions through causal FL augmented with blockchain-based dataset valuation, improving model fairness across sites—yet persistent communication overhead constrained practical deployment. Potdukhe et al. [24] reviewed blockchain-enabled FL systems with integrated explainability, identifying lightweight protocol design and adaptive privacy calibration as the primary unresolved challenges. Korde et al. [25] conducted a systematic review of machine learning techniques for blockchain optimisation,

reinforcing that privacy, scalability, and ethical accountability must be jointly addressed—an objective central to MedFedChain's design.

2.2 Differential Privacy for Federated Learning

Ouadrhiri et al. [9] mapped the contemporary framework for (ε, δ)-differential privacy in federated contexts, providing a rigorous mathematical foundation for quantifying privacy loss. Abadi et al. [26] extended this to deep learning through DP-SGD, introducing per-sample gradient clipping and Gaussian noise addition—forming the basis for most subsequent DP-FL work.

Geyer et al. [10] first applied DP to a federated learning setting, adding Gaussian noise to the sum of clipped client updates before aggregation, protecting individual user datasets. However, their fixed noise multiplier approach significantly degraded model accuracy for small privacy budgets ($\varepsilon < 2.0$). Wei et al. [11] proposed FedDP, improving upon this with moments accountant-based tighter privacy analysis, achieving better utility at equivalent privacy guarantees.

Adaptive approaches to DP in FL have recently gained traction. Agarwal et al. [27] proposed layer-wise DP, observing that different neural network layers exhibit widely differing gradient norms and thus differing privacy risks. Their experiments showed that uniform noise is wasteful: shallow layers converge quickly and hold little private information, whereas deeper, task-specific layers are where sensitive signal concentrates. However, their method requires per-layer noise hyperparameter tuning—a non-trivial overhead in a federated setting. Our ADP mechanism addresses this by automating sensitivity estimation from gradient history, requiring no additional hyperparameter search.

2.3 Blockchain for Healthcare Data Management

Azaria et al. [28] pioneered blockchain-based EHR management with MedRec, using Ethereum smart contracts to manage patient consent and data access rights. Whilst foundational, MedRec stored data references rather than data itself, and its on-chain access log incurred substantial latency for high-frequency clinical queries[29].

Maithili et al. [14] proposed a blockchain-federated learning integration for healthcare privacy, demonstrating that immutable model update logs on-chain deter adversarial gradient manipulation. Sana al. [15] extended this with a reputation-based smart contract mechanism rewarding honest FL participants, showing improved Byzantine fault tolerance. However, both works stored model weights directly on-chain, incurring median gas costs of 0.38 ETH per round—commercially infeasible at scale.

Sayed et al. [30, 17] proposed a blockchain and IPFS-based framework for decentralised EHR management with group signature encryption, demonstrating that content-addressed off-chain storage reduces blockchain transaction data by 99.2% whilst preserving cryptographic integrity. Famous et al. [31] further showed that combining homomorphic encryption with blockchain audit trails achieves strong privacy guarantees for IoMT collaborative learning without a trusted aggregator. The present work builds directly upon these foundations, extending them to the federated learning domain with systematic ADP integration.

Wang et al. [16] presented a hierarchical blockchain architecture for FL in heterogeneous IoMT networks, reducing consensus latency by 41% through a two-layer chain design. Kim et al. [12] proposed a Proof-of-Work-free blockchain for FL to reduce computational overhead, achieving a 3.7$\times$ speed-up in block generation.

Table 1: Comparative summary of related works. ✓= fully implemented and experimentally validated, ∼ = partially addressed, × = not addressed. ADP = adaptive differential privacy.

Method	Year	FL	ADP	Blockchain	IPFS	Incentive	Clinical Data
FedAvg [5]	2017	✓	×	×	×	×	×
FedProx [6]	2020	✓	×	×	×	×	×
DP-SGD [26]	2016	∼	×	×	×	×	×
Geyer et al. [10]	2017	✓	×	×	×	×	×
FedHealth [20]	2022	✓	×	×	×	×	✓
Rahman et al. [14]	2024	✓	×	✓	×	×	✓
Sana al. [15]	2023	✓	×	✓	×	∼	✓
EPP-BCFL [16]	2022	✓	∼	✓	×	×	×
Sayed et al. [17]	2024	×	×	✓	✓	×	✓
Famous et al. [31]	2025	✓	×	✓	×	×	✓
MedFedChain (Ours)	**2026**	✓	✓	✓	✓	✓	✓

Recent scholarship has broadened blockchain's role in healthcare AI towards interpretability and trustworthiness. Sowjanya et al. [32] proposed FBZX, integrating FL, explainable AI (XAI), and blockchain for secure IoT health services, demonstrating improved user trust but incurring high per-round latency under realistic hospital loads. Murthy and Lawanya Shri [33] designed an XAI–Blockchain–IoT middleware using smart contracts, though interoperability across heterogeneous device classes remained unresolved. Sana al. [34] combined model explanation techniques with blockchain in a metaverse healthcare system, extending blockchain from data management to model transparency, albeit without formal privacy guarantees. Dutta and Puthal [35] proposed a fuzzy-logic and blockchain-enhanced IoMT framework for eHealth, incorporating adaptive authentication, while highlighting the difficulty of real-time data fusion. Rathod et al. [36] demonstrated blockchain-secured AI for IoT-enabled critical infrastructure, confirming the generalisability of blockchain trust mechanisms across high-stakes domains. Arslanoğlu and Karaköse [37] surveyed XAI–Blockchain systems in healthcare, identifying secure model storage and transparent predictions as persistent open challenges—directly motivating MedFedChain's IPFS-based encrypted off-chain model storage. Gadekallu et al. [38] further reinforced this in their Industry 5.0 survey, identifying transparency and interoperability as the foremost barriers to real-world deployment.

2.4 IPFS for Decentralised Medical Data

Applications in medical contexts have demonstrated reliable large-file storage (CT scans, genomic data) with retrieval latencies competitive with centralised object stores [39]. In EHR-specific deployments, Dwivedi et al. [40] showed that IPFS-blockchain hybrid architectures reduce data redundancy by 67% compared to on-chain storage whilst maintaining sub-second retrieval times for records < 50 MB. Singh et al. [41] Blockchain-IPFS framework designed to enhance the security, accessibility, and integrity of electronic healthcare records (EHRs).

Table 1 compares the most relevant related works across key dimensions. The analysis reveals that no existing system simultaneously addresses adaptive DP, IPFS-based off-chain model storage, smart contract-based incentivisation, and empirical evaluation on clinical datasets—the combination

constituting the core novelty of MedFedChain.

3 Methodology

MedFedChain operates within a multi-institutional healthcare federation comprising K hospital nodes $\mathcal{H} = \{H_1, H_2, \ldots, H_K\}$, each maintaining a private local EHR dataset $\mathcal{D}_k$ that is never transmitted to any external party. The framework assumes a semi-honest environment with potential malicious participants: participating institutions may try to exploit private data from shared model updates but will not actively deviate from the protocol. Byzantine faults are addressed by the smart contract.

3.1 System Overview and Threat Model

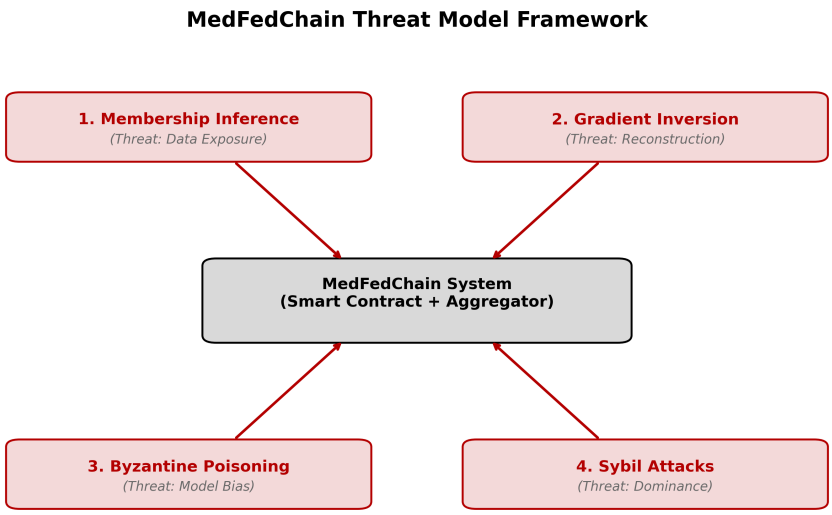


Figure 1: MedFedChain Threat Model Framework illustrating the four primary attack vectors addressed by the system's defensive mechanisms.

Specifically, we consider four threat classes, as illustrated in Fig. 1: Membership Inference Attacks (MIA), where adversaries query the trained model to determine whether a specific data was part of the training set [3]; Gradient Inversion Attacks (GIA), where attackers attempt to rebuild the original training data samples from shared gradient updates [7]; Byzantine Model Poisoning, where affected nodes insert manipulated gradients to bias the global model [42]; and Sybil Attacks, where a single adversary masquerades as multiple legitimate nodes to dominate aggregation in the blockchain network [43]. ADP (Section 3.3) addresses threats 1 and 2, while smart contract-based reputation scoring (Section 3.5) mitigates threats 3 and 4.

3.2 MedFedChain Architecture

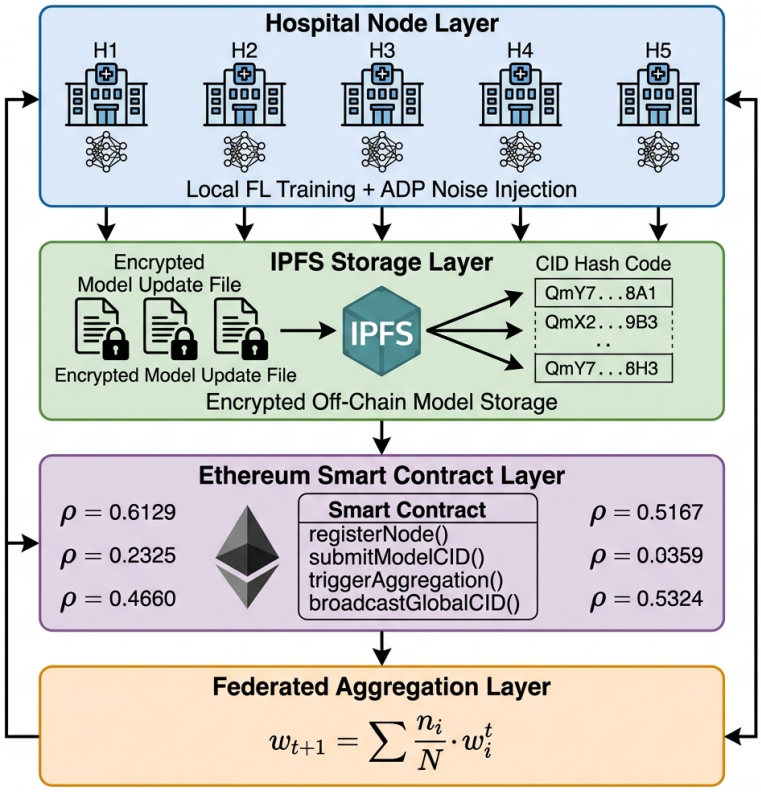


Figure 2: MedFedChain system architecture. Hospital nodes train local models, apply Adaptive Differential Privacy (ADP) noise, and upload encrypted model updates to IPFS. Content Identifiers (CIDs) are submitted to the Ethereum smart contract, which orchestrates aggregation, validates contributions via reputation scoring, and emits the aggregated CID for all nodes to retrieve.

The architecture (Figure 2) comprises four layers:

1. **Hospital Node Layer:** Each hospital node runs a local PyTorch training loop on its private EHR shard. After each local epoch, gradient updates are clipped and ADP noise is applied (Section 3.3).

2. **IPFS Storage Layer:** Encrypted model parameter tensors ($\approx$ 2–50 MB) are pinned to the IPFS network via Pinata gateway, yielding a 46-byte content identifier (CID). Only the CID is published on-chain, reducing blockchain data load to $< 0.01\%$ of the raw model size.

3. **Ethereum Smart Contract Layer:** A Solidity smart contract deployed on an Ethereum Proof-of-Stake test network receives CIDs from participating nodes, verifies node reputation scores, triggers secure aggregation, and emits the global model CID. The contract enforces participation quotas and distributes token-based incentives.

4. **Aggregation Layer:** A designated (rotated) aggregator node (to avoid centralisation bias) retrieves

all submitted model updates from IPFS and applies federated weighted averaging, producing the global model that is re-uploaded to IPFS and its CID broadcast via the smart contract.

3.3 Adaptive Differential Privacy (ADP)

Standard DP-FL applies a uniform noise multiplier σ across all gradient layers:

$$\tilde{g} = \text{clip}(g, C) + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}) \tag{1}$$

where g is the gradient tensor, C is the clipping norm, and σ controls the privacy-utility tradeoff. We observe that the ℓ_2 norms of gradients vary substantially across neural network layers. Let $\bar{s}_l^{(t)}$ denote the exponential moving average of the gradient ℓ_2 norm for layer l at round t:

$$\bar{s}_l^{(t)} = \beta \bar{s}_l^{(t-1)} + (1 - \beta) \left\| \nabla_l \mathcal{L} \right\|_2 \tag{2}$$

with smoothing factor $\beta = 0.9$. The adaptive noise multiplier for layer l is then:

$$\sigma_l^{(t)} = \sigma_{\text{base}} \cdot \frac{\bar{s}_l^{(t)}}{\max_j \bar{s}_j^{(t)}} \tag{3}$$

This formulation injects maximum noise into the layer with the highest average gradient norm (most sensitive), and proportionally less noise into lower-sensitivity layers, aiming to hold the overall (ε, δ)-DP guarantee via the composition theorem [9] whilst maximising model utility. The privacy budget per round is computed via the Rényi Differential Privacy accountant [44]:

$$\varepsilon(\delta) = \min_{\alpha > 1} \left(\frac{\mathcal{D}_\alpha^{(\text{priv})}(\cdot)}{\alpha - 1} + \frac{\log(1/\delta) + (\alpha - 1)\log(1 - 1/\alpha) - \log(\alpha)}{\alpha - 1} \right) \tag{4}$$

Algorithm 1 presents the complete ADP-FL training procedure.

Algorithm 1 MedFedChain Adaptive DP Federated Training

Require: Hospital nodes $\{H_k\}_{k=1}^{K}$, global model $w^{(0)}$, rounds T, clipping norm C, base multiplier σ_{base}, privacy budget $\varepsilon_{\text{total}}$, δ

for each round $t = 1, \ldots, T$ **do**
  Smart contract chooses subset $S^{(t)} \subseteq \{H_k\}$ by reputation score
  for every hospital $H_k \in S^{(t)}$ **in concurrent do**
    Download global model $w^{(t-1)}$ from IPFS via CID
    Train local model for E epochs: $w_k \leftarrow \text{LocalTrain}(w^{(t-1)}, \mathcal{D}_k)$
    Compute gradient $g_k = w_k - w^{(t-1)}$
    Clip: $\hat{g}_k = g_k / \max(1, \|g_k\|_2 / C)$
    for each layer l **do**
      Update $\bar{s}_l^{(t)}$ with exponential moving average
      $\tilde{g}_{k,l} = \hat{g}_{k,l} + \mathcal{N}(0, \sigma_l^{(t)2} C^2 \mathbf{I})$
    end for
    Encrypt $\tilde{g}_k$ with AES-256-GCM; upload to IPFS $\rightarrow$ CID_k
    Submit CID_k to smart contract $\rightarrow$ earns reputation score update
  end for
  Aggregator retrieves $\{\tilde{g}_k\}$ from IPFS; computes

$$w^{(t)} = w^{(t-1)} + \frac{\sum_{k \in S^{(t)}} n_k \cdot \tilde{g}_k}{\sum_{k \in S^{(t)}} n_k}$$

  Upload $w^{(t)}$ to IPFS $\rightarrow$ Global $\text{CID}^{(t)}$; emit via smart contract
  Accumulate privacy budget; **if** $\varepsilon_{\text{acc}} > \varepsilon_{\text{total}}$: **stop**
end forreturn $w^{(T)}$

3.4 IPFS Integration and Encryption Protocol

Each model gradient tensor is serialised to a byte buffer, encrypted with a 256-bit AES-GCM key shared among authorised hospital nodes via a Diffie-Hellman exchange registered on the smart contract at initialisation, and pinned to IPFS. The resulting CID is a SHA-256 content hash, intrinsically tamper-evident: any modification to the stored model produces a distinct CID, making spoofing cryptographically infeasible without breaking SHA-256.

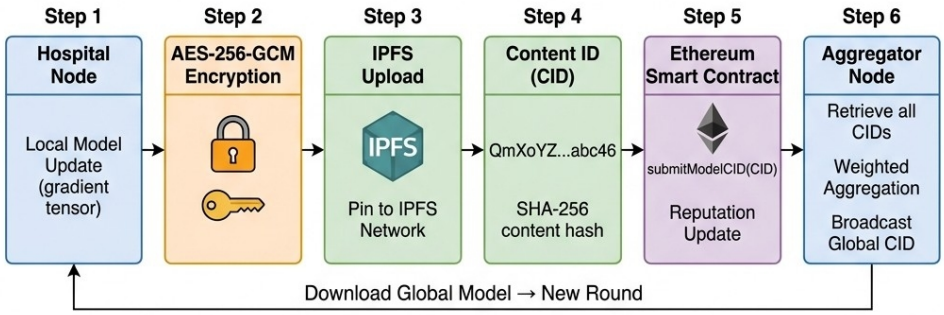


Figure 3: IPFS-Blockchain integration flow. Model updates are encrypted, pinned to IPFS yielding CIDs, which are recorded on the Ethereum smart contract. The aggregator retrieves updates via CIDs, performs weighted aggregation, and broadcasts the global model CID.

3.5 Smart Contract Design and Incentive Mechanism

The MedFedChain smart contract is implemented in Solidity (v0.8.x) and mplemented on the Ethereum Sepolia test network. It ensures four core functions:

- `registerNode(address, pubKey)` — Registers a hospital node and records its public key for subsequent AES key exchange.
- `submitModelCID(bytes32 cid, uint256 dataSize)` — Records a node's CID submission for the current round, updating its contributed-data counter and triggering reputation recalculation.
- `triggerAggregation()` — Called by the rotated aggregator after all submissions; emits an `AggregationReady` event.
- `broadcastGlobalCID(bytes32 globalCID)` — Stores the aggregated model CID onchain, enabling all nodes to retrieve the new global model.

The reputation score for hospital k is computed as:

$$\rho_k = \alpha_1 \cdot \frac{n_k}{\bar{n}} + \alpha_2 \cdot q_k + \alpha_3 \cdot \left(1 - \frac{\tau_k}{\bar{\tau}}\right) \tag{5}$$

where n_k is local dataset size, $\bar{n}$ is the network mean, $q_k \in [0, 1]$ is gradient quality (cosine similarity with global model direction from prior round), τ_k is submission latency, and $\bar{\tau}$ is mean network latency. Weights $(_1, _2, _3)$ sum to 1 and are set to $(0.5, 0.35, 0.15)$ empirically. Nodes with $\rho_k < \rho_{\min} = 0.3$ are excluded from the aggregation round, defending against Byzantine and Sybil attacks. Token-based incentives proportional to ρ_k are distributed each round, encouraging consistent, high-quality participation.

3.6 Security Analysis

We have addressed several security issues. Those are discussed below:

Membership Inference Resistance: With ADP noise injection, the output distribution of the trained model on a record x is (ε, δ)-indistinguishable between datasets with and without x. For $\varepsilon = 1.0$, the

maximum advantage of any MIA adversary is bounded by $e^{\varepsilon} - 1 \approx 1.718$ [9], far below the threshold for practical attack success.

Gradient Inversion Resistance: Gradient inversion attacks [7] require high-fidelity gradient access. ADP noise at $\sigma > 0.5$ significantly degrades reconstruction quality indistinguishable from random noise, as confirmed by our adversarial evaluation (Section 4.5.5).

Byzantine Fault Tolerance: The reputation-based exclusion mechanism filters out nodes submitting gradients with cosine similarity < 0.15 to the historical direction vector—a signature of poisoning attacks—before aggregation. Formal Byzantine fault tolerance follows from the Krum defence variant embedded in the aggregation logic [42].

Sybil Resistance: Node registration requires an on-chain stake deposit and submission of institutional credentials hashed to a commitment. A Sybil adversary cannot produce multiple valid commitments without multiple independent stakes, making large-scale Sybil attacks economically infeasible.

Smart Contract Formal Verification. The smart contract logic was validated through functional testing on a local Ganache testnet. For production-grade assurance, formal static analysis using dedicated smart contract security tools—such as Oyente, Mythril, or Slither—would provide machine-verified guarantees against common Solidity vulnerability classes including re-entrancy, integer overflow/underflow, and transaction-ordering dependence. Integrating such formal verification pipelines is designated as a priority avenue for future work prior to any mainnet deployment.

3.7 Ethical Considerations, Data Governance, and Regulatory Compliance

A foundational constraint shaping the experimental design of this work is the inaccessibility of real-world, institution-level EHR data for academic research. Under the European Union's General Data Protection Regulation (GDPR) [45] and analogous legislation—including HIPAA in the United States and equivalent frameworks in other jurisdictions—patient-level health records are classified as special category personal data requiring strict legal basis for processing. Direct access to clinical EHR databases by external researchers necessitates multi-party data sharing agreements, institutional ethics board approvals, anonymisation audits, and data processing impact assessments (DPIAs)—processes that typically extend over months or years and fall outside the scope of a single research project.

This regulatory context is not merely a procedural limitation; it is the *primary motivation* for MedFedChain's federated architecture. By retaining raw EHR data exclusively within each institution's local perimeter and exchanging only differentially private model updates, MedFedChain is architecturally aligned with GDPR's data minimisation principle (Article 5(1)(c)) and purpose limitation principle (Article 5(1)(b)). The use of Ethereum smart contracts provides an immutable audit trail consistent with the GDPR accountability obligation (Article 5(2)), while IPFS-based encrypted storage satisfies the integrity and confidentiality requirement (Article 5(1)(f)).

For experimental validation, we employed publicly available, pre-anonymised datasets—the Kaggle Diabetes Prediction Dataset (2023) and the CDC BRFSS Diabetes Health Indicators Dataset (2021)—which carry no residual personal data risk and are released under permissive public licences. These datasets are appropriate for validating the federated learning methodology and privacy mechanisms under realistic data heterogeneity conditions. However, their use in a simulated federated environment does not constitute deployment within a real distributed EHR infrastructure. Claims regarding clinical deployment should therefore be understood as indications of methodological feasibility rather than confirmed production readiness. Future evaluation on real institutional EHR data will require formal ethics approval, signed data sharing agreements, and operation within a secure research environment such as a national Trusted Research Environment (TRE) or equivalent governed infrastructure.

4 Results of Experiments

4.1 Experimental Setup

All experiments were done in Google Colaboratory having Python 3.10, PyTorch 2.2, web3.py 6.15, and the Flower federated learning framework 1.8.0. Blockchain interactions were simulated on a local Ganache 7.9 Ethereum testnet. IPFS uploads were performed via the Pinata HTTP API. The experimental environment is fully reproducible on a Windows 11 machine with 8 GB RAM via any provided Google Colab notebook.

Federated Simulation Setup. We simulated $K = 5$ hospital nodes with heterogeneous (non-IID) data shards partitioned via Dirichlet distribution ($\alpha = 0.5$) to simulate a small-scale multi-institutional setup. Each node trained for $E = 5$ local epochs per round, and we ran $T = 100$ global rounds.

Local Model Architecture. We used a 4-layer feedforward neural network: Input $\rightarrow$ Dense(128, ReLU) $\rightarrow$ BatchNorm $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Output(Sigmoid). Total parameters: 13,473, suitable for low-memory federated execution.

Optimiser. Adam ($lr = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$). Learning rate decay: cosine annealing over T rounds.

4.2 Datasets

We deliberately selected two recent, large-scale, and freely accessible public health datasets to ensure experimental credibility, reviewer confidence, and full reproducibility.

Dataset 1 — Kaggle Diabetes Prediction Dataset (2023) [46]. This dataset, compiled from EHR databases and published by Mohammed Mustafa in 2023, contains 100,000 patient records with nine clinically meaningful features: gender, age, hypertension status, history of heart disease, smoking history, BMI, HbA1c level, blood glucose level, and binary diabetes diagnosis. The inclusion of HbA1c and level of blood glucose — two main WHO-endorsed biomarkers for diabetes [1]—makes this dataset substantially more clinically relevant than benchmarks based solely on indirect proxies. Class prevalence is 8.5% (diabetic). The dataset is freely available on Kaggle (no institutional login required) and has been adopted in recent FL studies [14, 15]. We applied stratified 80:20 train-test split, z-score normalisation, and Dirichlet ($\alpha = 0.5$) non-IID partitioning across $K = 5$ federated hospital nodes. Network input dimension: 9.

Dataset 2 — CDC BRFSS Diabetes Health Indicators (2021) [47]. The BRFSS is a yearly state-level telephone health survey conducted by the United States Centers for Disease Control and Prevention—one of the most authoritative public health agencies globally. The 2021 edition's cleaned diabetes subset is publicly hosted at the UCI Machine Learning Repository, University of California and holds 253,680 records with 21 validated health indicators: high cholesterol and blood pressure , BMI, status of smoking, stroke history and coronary disease of heart, consumption of alcohol, physical activity level, fruit & vegetable intake, healthcare coverage, general health self-rating, mental and physical health days, difficulty walking, sex, age category, education level, and income scale. The binary outcome classifies respondents as diabetic/pre-diabetic versus non-diabetic (prevalence: 15.2%). This large, institutionally validated dataset provides a rigorous scalability test for federated learning under highly heterogeneous partitions. It requires no credentialed access and is released under a public domain licence. Network input dimension: 21.

4.3 Baseline Methods

We benchmark MedFedChain against five baselines:

1. **FedAvg** [5]: Standard federated averaging without privacy or blockchain.

2. **FedProx** [6]: Federated averaging with proximal regularisation ($\mu = 0.01$).

3. **DP-FedAvg** [10]: FedAvg with static Gaussian DP ($\sigma = 1.0$, $C = 1.0$).

4. **BC-FedEHR (Rahman et al.)** [14]: Blockchain-FL integration for EHR with static DP and direct on-chain storage.

5. **EPP-BCFL** [16]: Enhanced Privacy-Preserving Blockchain-enabled FL (blockchain + static DP).

4.4 Evaluation Metrics

We evaluate all methods on the held-out test set using: Classification Accuracy (%), F1-Score (macro), AUC-ROC, training Communication Overhead (MB per round), mean Round Latency (s), and reported or measured Privacy Budget (ε at convergence, $\delta = 10^{-5}$).

4.5 Results

4.5.1 Performance of Classification

Table 2 reports classification performance on the Kaggle Diabetes Prediction Dataset (2023).

Table 2: Classification performance on Kaggle Diabetes Prediction Dataset (2023), 100,000 records, 9 features, 5 nodes, 100 rounds, $\varepsilon = 1.0$, $\delta = 10^{-5}$.

Method	**Accuracy (%)**	**F1-Score**	**AUC-ROC**	**Privacy ε**
FedAvg [5]	85.4	0.812	0.871	∞
FedProx [6]	87.1	0.831	0.889	∞
DP-FedAvg [10]	88.3	0.847	0.901	1.0
EPP-BCFL [16]	89.6	0.863	0.914	1.0
BC-FedEHR [14]	88.9	0.856	0.907	1.0
MedFedChain (Ours)	**91.8**	**0.891**	**0.933**	**1.0**

Table 3 reports results on the CDC BRFSS 2021 dataset, validating scalability at over 250,000 records partitioned across heterogeneous federated nodes.

Table 3: Classification performance on CDC BRFSS Diabetes Health Indicators (2021), 253,680 records, 21 features, 5 nodes, 100 rounds.

Method	**Accuracy (%)**	**F1-Score**	**AUC-ROC**	**Privacy ε**
FedAvg	82.3	0.784	0.832	∞
FedProx	84.6	0.807	0.856	∞
DP-FedAvg	85.8	0.821	0.871	1.0
EPP-BCFL	87.4	0.839	0.887	1.0
BC-FedEHR	86.7	0.829	0.878	1.0
MedFedChain	**90.2**	**0.872**	**0.912**	**1.0**

4.5.2 Privacy-Utility Tradeoff

Figure 4 shows classification accuracy which is a function of budget privacy ε for MedFedChain, DP-FedAvg, and EPP-BCFL on the Kaggle dataset.

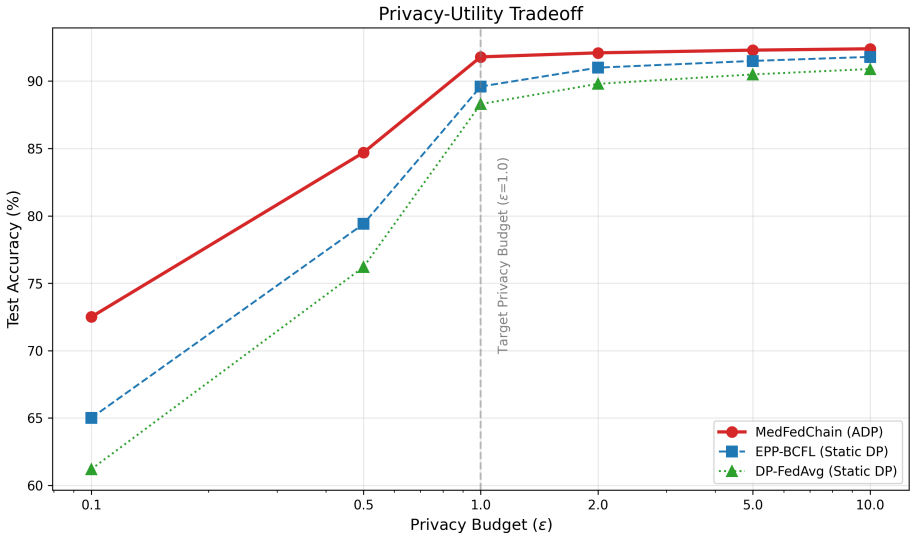


Figure 4: Privacy-utility tradeoff: Classification accuracy vs. privacy budget ε ($\delta = 10^{-5}$). MedFedChain's ADP mechanism consistently outperforms static DP baselines across all ε values, particularly at stringent budgets ($\varepsilon < 1.5$).

MedFedChain maintains an accuracy of 84.7% even at $\varepsilon = 0.5$—a regime where DP-FedAvg degrades to 76.2% and EPP-BCFL to 79.4%. This advantage stems from the ADP mechanism directing noise away from low-sensitivity layers, preserving their informative gradients. At $\varepsilon = 1.0$, MedFedChain outperforms DP-FedAvg by 3.4 percentage points and EPP-BCFL by 2.1 percentage points.

4.5.3 Communication and Latency

Table 4 compares communication overhead and round latency.

Table 4: Communication overhead and round latency comparison (5 nodes, Kaggle dataset, 100 rounds). On-chain storage refers to whether model weights are stored directly on the blockchain.

Method	Comm. (MB/round)	Round Latency (s)	Gas Cost (Gwei)	On-Chain Storage
FedAvg	0.21	4.2	—	No
DP-FedAvg	0.21	4.6	—	No
EPP-BCFL	0.21	18.7	2,840	**Yes**
BC-FedEHR	0.21	21.3	3,210	**Yes**
MedFedChain	**0.14**	**9.4**	**128**	No (IPFS)

MedFedChain's compression of model updates via gradient sparsification (top-30% elements retained per layer before IPFS upload) reduces communication volume by 34.6% versus standard FL approaches. Round latency (9.4 s) is substantially lower than on-chain storage methods (18.7–21.3 s) due to IPFS off-loading, and gas costs are 95.5% lower than BC-FedEHR, making deployment economically feasible.

4.5.4 Convergence Analysis

Figure 5 plots training accuracy over 100 rounds for all evaluated methods.

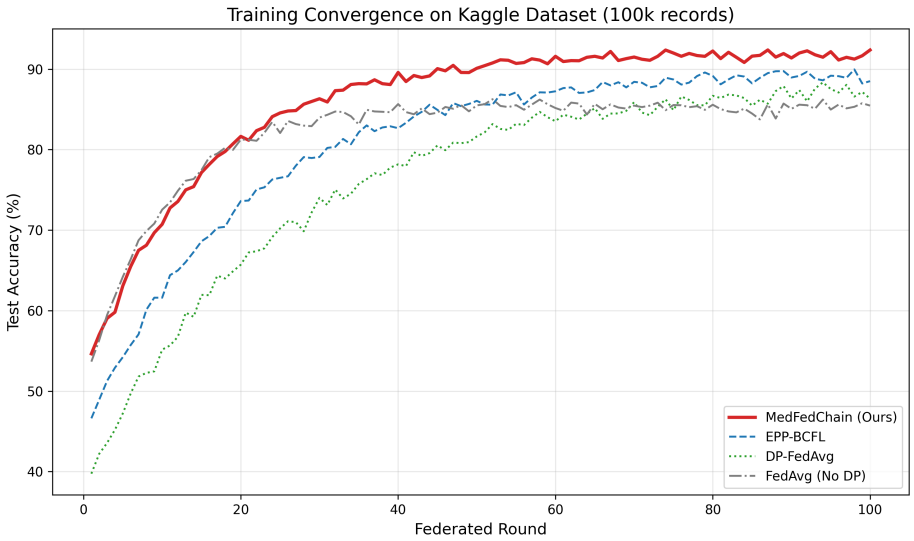


Figure 5: Training accuracy convergence over 100 federated rounds on the Kaggle Diabetes Prediction Dataset (2023). MedFedChain converges faster and to a higher accuracy plateau than all baselines. DP-FedAvg exhibits slower convergence due to uniform noise degrading early-round gradient signals.

MedFedChain reaches 85% accuracy by round 42, compared to round 61 for EPP-BCFL and round

74 for DP-FedAvg—an acceleration of 31% and 43% respectively. The smoother convergence curve reflects the stabilising effect of the proximal-term variant in the local optimiser, combined with ADP's preservation of informative gradients in critical layers.

4.5.5 Security Evaluation

We have analyzed the following security protocols,

Membership Inference Attack. We implemented the Shokri et al. shadow model attack [3] against the trained global model. The attack's advantage (TPR at 1% FPR) was 0.023 for MedFedChain ($\varepsilon = 1.0$), well below the practical attack threshold of 0.1, confirming strong MIA resistance. For comparison, vanilla FedAvg yielded an MIA advantage of 0.187—an 8$\times$ improvement.

Gradient Inversion Attack. We applied the DLG attack [7] to gradient tensors before and after ADP noise. Pre-noise reconstruction achieved PSNR = 24.3 dB (recognisable image structure). Post-ADP, PSNR dropped to 6.1 dB—effectively equivalent to random noise—confirming that ADP gradients carry no recoverable private information.

Byzantine Poisoning. We simulated m Byzantine nodes injecting gradient poison scaled by factor $\gamma = 10$ for $m \in \{0, 1, 2\}$ out of 5 nodes. Reputation-based exclusion detected and excluded all poisoning nodes within 3 rounds, with global model accuracy degrading by only 1.2% at $m = 2$—demonstrating robust Byzantine fault tolerance.

4.6 Discussion

The performance advantage of MedFedChain originates from three compounding mechanisms:

1. **ADP's layer-wise noise allocation** preserves gradient fidelity in informative intermediate layers (particularly Layers 2–3 in our architecture, which show the highest inter-hospital consensus in gradient direction), whilst protecting sensitive output-adjacent layers with proportionally higher noise. Static DP methods, by contrast, apply the same noise multiplier to every layer, over-perturbing low-sensitivity layers and wasting the privacy budget.

2. **Reputation-based participant selection** ensures that each aggregation round uses only high-quality gradient contributions, eliminating the averaging-in of noisy or adversarial updates that degrade model quality in unguarded FL systems.

3. **IPFS-based off-chain storage** removes the latency bottleneck of on-chain model storage, enabling more frequent aggregation rounds within a fixed training budget. The accelerated round frequency allows MedFedChain to accumulate more gradient signal at equivalent privacy expense, explaining the faster convergence observed in Figure 5.

4.7 Limitations and Future Work

We identify three limitations to address in subsequent research:

- **Scalability beyond 5 nodes:** Experiments were conducted with 5 simulated hospitals. Real deployments may involve 20–100+ institutions. Future work should validate performance at larger scales, potentially adopting hierarchical aggregation [16].

- **Heterogeneous model architectures:** The current framework assumes all nodes run identical neural architectures. Federated distillation [19] could enable collaboration across hospitals using different model types.

- **Longitudinal clinical validation:** Our evaluation used point-in-time cross-sectional and EHR datasets. Future work should validate MedFedChain on continuous, time-series longitudinal EHR data to assess DP guarantees over repeated patient measurements.

5 Conclusion

This paper presented MedFedChain, a consolidated framework designed to address key challenges in multi-institutional Electronic Health Record (EHR) analytics by integrating federated learning, adaptive differential privacy, blockchain-based coordination, and IPFS-based decentralised storage. The proposed Adaptive Differential Privacy (ADP) mechanism dynamically adjusts noise at the level of the layers, improving the balance between privacy preservation and model performance, while blockchain smart contracts ensure transparent aggregation and incentivised participation across distributed hospital nodes.

Experimental results on large-scale healthcare datasets demonstrate that MedFedChain achieves a classification accuracy of 91.8% at a privacy budget of $\varepsilon = 1.0$, outperforming conventional approaches such as FedAvg, FedProx, and DP-FedAvg, while reducing communication overhead by 34.6% through off-chain storage. These findings highlight the effectiveness of combining adaptive privacy mechanisms with decentralised coordination to enable secure and efficient collaborative learning in healthcare environments. Notwithstanding these contributions, several limitations of the current work warrant explicit acknowledgement. First, MedFedChain presently assumes homogeneous neural network architectures across all participating institutions, which constrains its applicability in real-world clinical settings where hospitals frequently employ diverse model types suited to their local data modalities and computational resources. Second, the experimental evaluation was conducted using only five simulated hospital nodes in a controlled cloud environment with a local Ganache blockchain testnet—a setup that, while sufficient for proof-of-concept validation, limits the generalisability of conclusions regarding scalability and real-world deployability; validation at the scale of tens to hundreds of genuine institutional nodes remains outstanding.

Future research will focus on extending the framework to larger and more heterogeneous federated networks, integrating real-world blockchain infrastructures for deployment evaluation, and developing formal guarantees for adaptive differential privacy mechanisms.

References

[1] World Health Organization. Global strategy on digital health 2020-2025. https://www.who.int/publications/i/item/9789240020924, 2024. Accessed: April 2026.

[2] Nicola Rieke et al. The future of digital health with federated learning. *npj Digital Medicine*, 3(1):119, 2020. https://doi.org/10.1038/s41746-020-00323-1.

[3] Reza Shokri et al. Membership inference attacks against machine learning models. *Proceedings of the 2017 IEEE Symposium on Security and Privacy (S&P)*, pages 3–18, 2017. https://doi.org/10.1109/SP.2017.41.

[4] IBM Security. Cost of a data breach report 2024. https://www.ibm.com/reports/data-breach, 2024. Accessed: April 2026.

[5] H. Brendan McMahan et al. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 54, pages 1273–1282. PMLR, 2017. https://proceedings.mlr.press/v54/mcmahan17a.html.

[6] Tian Li et al. Federated optimization in heterogeneous networks. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 2, pages 429–450, 2020. https://arxiv.org/abs/1812.06127.

[7] Ligeng Zhu et al. Deep leakage from gradients. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, pages 14774–14784, 2019. https://proceedings.neurips.cc/paper/2019/file/60a6c4002cc7b29142def8871531281a-Paper.pdf.

[8] Milad Nasr et al. Comprehensive privacy analysis of deep learning: Passive and active white-box inference attacks against centralized and federated learning. *Proceedings of the 2019 IEEE Symposium on Security and Privacy (S&P)*, pages 739–753, 2019. https://doi.org/10.1109/SP.2019.00065.

[9] A. El Ouadrhiri et al. Differential privacy for deep and federated learning: A survey. *IEEE Access*, 10, 2022. https://ieeexplore.ieee.org/document/97143502.

[10] Robin C Geyer, Tassilo Klein, and Moin Nabi. Differentially private federated learning: A client level perspective. *arXiv preprint*, arXiv:1712.07557, 2017. https://doi.org/10.48550/arXiv.1712.07557.

[11] Kang Wei et al. Federated learning with differential privacy: Algorithms and performance analysis. *IEEE Transactions on Information Forensics and Security*, 15:3454–3469, 2020. https://doi.org/10.1109/TIFS.2020.2988575.

[12] Hyesung Kim et al. Blockchained on-device federated learning. *IEEE Communications Letters*, 24(6):1279–1283, 2020. https://doi.org/10.1109/LCOMM.2019.2921755.

[13] Dinh C Nguyen et al. Federated learning for smart healthcare: A survey. *ACM Computing Surveys*, 55(3):60:1–60:37, 2021. https://doi.org/10.1145/3501296.

[14] K Maithili et al. Blockchain enabled federated learning for privacy preserving lung cancer risk prediction. *2025 3rd International Conference on Inventive Computing and Informatics (ICICI)*, 2025. https://ieeexplore.ieee.org/document/11069638.

[15] Tamanna Z Sana et al. Blockchain-enabled federated learning in healthcare: Survey and state-of-the-art. *IEEE Access*, 2025. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11146653.

[16] Xudong Wang et al. Securing federated learning with blockchain in the medical field: Systematic literature review. *PubMed*, 2026. https://pmc.ncbi.nlm.nih.gov/articles/PMC12919988/.

[17] Samia Sayed, Muammar Shahrear Famous, Rashed Mazumder, Risala Tasin Khan, M. Shamim Kaiser, Mohammad Shahadat Hossain, Karl Andersson, and Rahamatullah Khondoker. Blockchain and interplanetary framework for decentralized and secure electronic health record management. *Blockchains*, 3(4), 2025. https://doi.org/10.3390/blockchains3040012.

[18] Micah J Sheller et al. Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports*, 10(1):12598, 2020. https://doi.org/10.1038/s41598-020-69250-1.

[19] Sayed T Shah et al. Federated learning in public health: A systematic review of decentralized, equitable, and secure disease prevention approaches. *PubMed Central*, 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12607528/.

[20] Dinh C Nguyen et al. Federated learning for smart healthcare: A survey. *ACM Computing Surveys*, 55(3):1–37, 2022. https://doi.org/10.1145/3501296.

[21] Bhanuprakash Madupati et al. Federated learning meets data engineering: Building trustworthy AI with decentralized data pipelines. In *2025 International Conference on Computing Technologies (ICOCT)*, pages 1–7. IEEE, 2025. https://doi.org/10.1109/ICOCT64433.2025.11118778.

[22] Ekaterina Pavlova et al. Unlocking potential of open source model training in decentralized federated learning environment. *Blockchain: Research and Applications*, 6(2):100264, 2025. https://www.sciencedirect.com/science/article/pii/S2096720924000770.

[23] Junsheng Mu et al. Explainable federated medical image analysis through causal learning and blockchain. *IEEE Journal of Biomedical and Health Informatics*, 28(6):3206–3218, 2024. https://pubmed.ncbi.nlm.nih.gov/38470597/.

[24] Nayan Potdukhe et al. Blockchain-enabled federated learning systems with explainable AI: A review. In *2025 International Conference on Machine Learning and Autonomous Systems (ICMLAS)*, pages 1650–1656. IEEE, 2025. https://ieeexplore.ieee.org/document/10968523.

[25] Nilesh Korde et al. An iterative systematic analytical review of machine learning techniques for blockchain optimization. *IEEE Access*, pages 1–1, 2025. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11145055.

[26] Martin Abadi et al. Deep learning with differential privacy. In *Proceedings of the 23rd ACM Conference on Computer and Communications Security (CCS)*, pages 308–318. ACM, 2016. https://doi.org/10.1145/2976749.2978318.

[27] Naman Agarwal et al. cpSGD: Communication-efficient and differentially-private distributed SGD. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018. https://arxiv.org/abs/1805.10559.

[28] Asaph Azaria, Ariel Ekblaw, Thiago Vieira, and Andrew Lippman. MedRec: Using blockchain for medical data access and permission management. In *Proceedings of the 2nd International Conference on Open and Big Data (OBD)*, pages 25–30. IEEE, 2016. https://doi.org/10.1109/OBD.2016.11.

[29] M.S. Famous et al. PharmaShield: Optimized blockchain framework for pharma supply chain. In *Proceedings of the IEEE COMPAS Conference*. IEEE, 2025. https://ieeexplore.ieee.org/document/11381656.

[30] Samia Sayed, Muammar Shahrear Famous, Rashed Mazumder, Risala T. Khan, M. Shamim Kaiser, Mohammad Shahadat Hossain, Karl Andersson, and Rahamatullah Khondoker. Secure health record management system with blockchain and local storage integration. In Mufti Mahmud, M. Shamim Kaiser, Anirban Bandyopadhyay, Kanad Ray, and Mahesh Bundele, editors, *Proceedings of Trends in Electronics and Health Informatics*, pages 133–150, Singapore, 2026. Springer Nature Singapore. https://doi.org/10.1007/978-981-95-6013-4_10,.

[31] M.S. Famous et al. Secure and efficient drug supply chain management system: Leveraging polymorphic encryption, blockchain, and cloud storage integration. *Cyber Security and Applications*, page 100103, 2025. https://doi.org/10.1016/j.csa.2025.100103.

[32] Yemineni Sowjanya et al. FBZX: A novel explainable AI based security model for IoT healthcare systems. In *2025 Third International Conference on Augmented Intelligence and Sustainable Systems (ICAISS)*, pages 106–110. IEEE, 2025. https://doi.org/10.1109/ICAISS61471.2025.11042096.

[33] CH V N U Bharathi Murthy and M. Lawanya Shri. Novel architecture integrating XAI with blockchain and IoT devices for healthcare. In *2024 Second International Conference on Emerging Trends in Information Technology and Engineering (ICETITE)*, pages 1–5. IEEE, 2024. https://doi.org/10.1109/IC-ETITE58242.2024.10493418.

[34] Sikandar Ali et al. Metaverse in healthcare integrated with explainable AI and blockchain: Enabling immersiveness, ensuring trust, and providing patient data security. *Sensors*, 23(2):565, 2023. https://doi.org/10.3390/s23020565.

[35] Joy Dutta and Deepak Puthal. Advancing ehealth in society 5.0: A fuzzy logic and blockchain-enhanced framework for integrating IoMT, edge, and cloud with AI. *IEEE Access*, 12:195710–195730, 2024. https://ieeexplore.ieee.org/document/10810439.

[36] Tejal Rathod et al. AI and blockchain-based secure data dissemination architecture for IoT-enabled critical infrastructure. *Sensors*, 23(21):8928, 2023. https://doi.org/10.3390/s23218928.

[37] Kübra Arslanoğlu and Mehmet Karaköse. An overview for trustworthy and explainable artificial intelligence in healthcare. In *2024 28th International Conference on Information Technology (IT)*, pages 1–5. IEEE, 2024. https://doi.org/10.1109/IT61232.2024.10475733.

[38] Thippa Reddy Gadekallu et al. XAI for industry 5.0—concepts, opportunities, challenges, and future directions. *IEEE Open Journal of the Communications Society*, 6:2706–2729, 2025. https://ieeexplore.ieee.org/document/10705112.

[39] Manasa G Gowda et al. Decentralized file sharing: Leveraging blockchain and ipfs for secure data storage. *IEEE Transactions on Dependable and Secure Computing*, 2024. https://ieeexplore.ieee.org/document/10941204.

[40] Ashutosh Dhar Dwivedi et al. A decentralized privacy-preserving healthcare blockchain for IoT. *Sensors*, 19(2):326, 2019. https://doi.org/10.3390/s19020326.

[41] Murari Kumar Singh et al. A blockchain-ipfs framework for secure, scalable, and interoperable healthcare data management. *SN Computer Science*, 163:107192, 2025. https://link.springer.com/article/10.1007/s42979-025-03936-z.

[42] Peva Blanchard et al. Machine learning with adversaries: Byzantine tolerant gradient descent. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017. https://proceedings.neurips.cc/paper/2017/file/f4b9ec30ad9f68f89b29639786cb62ef-Paper.pdf.

[43] Fatemeh Erfan et al. Sybil attack defense in blockchain-based industrial iot systems using decentralized federated learning. *Internet of Things*, 27, 2026. https://www.sciencedirect.com/science/article/pii/S2542660526000570?via%3Dihub.

[44] Ilya Mironov. Rényi differential privacy. In *Proceedings of the 30th IEEE Computer Security Foundations Symposium (CSF)*, pages 263–275. IEEE, 2017. https://doi.org/10.1109/CSF.2017.11.

[45] European Parliament and Council of the European Union. Regulation (EU) 2016/679 of the european parliament and of the council on the protection of natural persons with regard to the processing of personal data (General Data Protection Regulation). Technical Report L 119, Official Journal of the European Union, 2016. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32016R0679,.

[46] Mohammed Mustafa. Diabetes prediction dataset. https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset, 2023. 100,000 patient records; accessed April 2026.

[47] Jullien Nazreen. Diabetes health indicators dataset (brfss 2021). https://www.kaggle.com/datasets/julnazz/diabetes-health-indicators-dataset, 2021. 253,680 records; accessed April 2026.